

Ernesto Rivera

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EDUCATION

Boston University

Boston, MA

B.A. in Computer Science

Jan. 2023 - May 2026

- Coursework: Data Structures, Algorithms, Distributed Systems, Full-Stack, Artificial Intelligence, Machine Learning, Operating Systems, Software Engineering, Computational Probability, Logic Design, AI Ethics

EXPERIENCE

Wayfair

Boston, MA

Software Engineer Intern

Jun. 2025 - Aug. 2025

- Resolved a critical **26%** customer drop-off in the WFCC application funnel by owning the end-to-end deployment of a server-side **PHP** feature, delivering a state-aware banner that guided users through a key conversion point.
- Architected and implemented a scalable E2E test framework using **Playwright** and the **Page Object Model (POM)**, creating custom fixtures to solve complex state management via browser cookie injection.
- Prevented a high-risk deployment targeting a **14%** conversion gap by mapping complex user flows to **Mermaid** diagrams, ensuring safer production updates in collaboration with Product and Marketing stakeholders.

Whiting-Turner Contracting Co.

Irvine, CA

Data Architecture Engineer

May 2024 - Oct. 2024

- Developed a scalable task management application by architecting the backend and optimizing system logic, reducing scheduling delays by **12%**, while creating a robust technical framework for junior interns to expand.
- Restructured a **1 billion+** data point database into job site-specific databases, using **Azure IoT Hub**, **Stream Analytics**, **Cosmos DB**, and **Blob Storage**, designed for reliable, real-time multi-tenant sessions at scale.

Global Scope Research Lab

Boston, MA

Research Assistant

Jan. 2024 - May 2024

- Analyzed a **32 GB**, 370 million-row dataset from the US Patent Directory to uncover innovation trends before and after Hurricane Katrina, shedding light on disaster-driven innovation dynamics.
- Collaborated with a Research Assistant using **NumPy** and **Dask** on a shared computing cluster, streamlining data processing to extract scalable insights into patent trends and support post-disaster policy analysis.

MyTA

Remote

Startup Founding Engineer

Jul. 2023 - Mar. 2024

- Engineered an educational chatbot by integrating **Meta AI's Nougat** for high-accuracy OCR and **Pinecone's** vectorization, collaborating in a team of 4 and liaising with **9 universities** to drive adoption.
- Refined chatbot functionality to process entire professorial materials via **OCR-to-MultiMarkdown** translation, enabling dynamic, personalized student support through **vectorized** content integration.

PROJECTS AND COMPETITIONS

ViewPrep

Nov. 2024 - Jan. 2025

- Built a real-time AI interview platform with bidirectional audio streaming via **WebSockets**, **OpenAI ASR & LLM**, and a **custom WebAudio pipeline** for high-fidelity capture.
- Designed and deployed a **scalable AWS architecture**, integrating **ECS** (Flask backend), **API Gateway**, **RDS** (PostgreSQL), and **CloudFront** (Next.js frontend) to support **multi-tenant** interview sessions.
- Engineered an optimized real-time audio streaming system with **custom AudioWorklet processors**, achieving **sub-second latency** and enabling immediate AI-driven feedback on speech patterns and response quality.

Gloo Hackathon: Best Technology - \$50,000 Prize

Oct. 2023

- Pioneered the development of personalized faith-based AI counselors, accessible via targeted social media and search engine ads, which significantly enhanced outreach efficiency and achieved a **100x** cost reduction.
- Engineered an AI-driven **MetaHuman** backend with **Inworld AI** and **Unreal Engine**, enabling immersive **3D** interaction that improved user engagement, leveraging human-to-LLM retention stats.

OCR Digit Machine Learning Classifier

Jun. 2024

- Developed an OCR application using a **custom K-Nearest Neighbors (KNN)** classifier to identify digits from **60,000 MNIST** images, achieving **95%** accuracy by optimizing preprocessing and feature extraction methods.
- Enhanced the application with a user-friendly **GUI** built using **Tkinter** and **PIL** in **Python**, utilizing ImageOps, ImageFilter, and Numpy for real-time digit recognition to demonstrate machine learning capabilities.

SKILLS

Technologies: Python, JavaScript, TypeScript, C++, C, Java, Go, PHP, OCaml, LangGraph, SQL, GraphQL, PyTorch, NumPy, Pandas, Dask, Pinecone, OCR (Nougat, MMOCR, DocTR), Next.js, React, Flask, FastAPI, Jest, Playwright, GCP, AWS, Docker, Terraform, DataDog, Kafka, Azure, Firebase, Git, Bash, CI/CD, Agile, Mermaid

Interests: Quantum Computing, AI, Robotics, Cooking, Film Photography, Running, Specialty Coffee